

Stage 17: Reviews, Ratings, UGC & Community Q&A Engine

1. Architectural Overview

The **Reviews, Ratings, UGC & Community Q&A Engine** establishes trustworthy social proof, verified purchaser recognition, multi-criteria ratings, media attachments, vendor interaction, and administrative moderation across the Alight International Multi-Vendor Marketplace.



2. Core Components & Database Structure

(`V43__reviews_ratings_and_qa_enhancements.sql`)

2.1 Customer Reviews (`reviews`)

- `product_id` & `user_id` : Enforces a strict one-review-per-customer constraint per product (`uq_product_user_review`).
- `rating` : 1 to 5 star rating scale.
- `is_verified_purchase` : Dynamically verified against customer order item fulfillment records.
- `images` : JSONB array storing uploaded UGC photo URLs.
- `status` : Moderation lifecycle (`APPROVED` , `PENDING_MODERATION` , `REJECTED` , `FLAGGED_SPAM`).
- `helpful_count` & `unhelpful_count` : Aggregated community voting totals.
- `vendor_response` & `vendor_responded_at` : Official store response thread.

2.2 Review Helpfulness Voting (`review_votes`)

- Tracks unique user votes per review (`uq_review_user_vote`). Supports switching between `HELPFUL` and `UNHELPFUL` , as well as toggling off.

2.3 Community Q&A Threads (`product_questions` & `product_answers`)

- Customers can post pre-purchase questions (`product_questions`).
- Answers (`product_answers`) can be posted by vendors, platform admins, or verified buyers.
- Vendors and verified sellers automatically receive official badges (`is_verified_seller = true` , `is_accepted = true`).

3. Product Rating Aggregation Engine

Whenever a review is created, updated, or moderated:

1. `ReviewServiceImpl.updateProductRatingCache(productId)` executes:

- Fetches all `APPROVED` reviews for the product.

- Calculates the exact arithmetic mean rounded to 2 decimal places:

$$\text{Average Rating} = \frac{\sum \text{Ratings}}{\text{Total Approved Reviews}}$$

- Atomically updates `products.average_rating` and `products.review_count` .

4. Verification & Integration Suite

Verified via `ReviewsRatingsAndQaIntegrationTest.java` :

✅ **Verified Purchase Recognition** : Verified buyers receive badges while unverified users can post normal reviews.

✅ **Product Rating Cache Sync** : Automatic recalculation of `products.average_rating` and `products.review_count` .

- ✅ **Rating Statistics Breakdown:** 1-5 star distributions, percentages, verified counts, and UGC photo counts.
- ✅ **Helpful/Unhelpful Voting:** Adding, switching, and untoggling votes.
- ✅ **Vendor Official Replies:** Store reply thread with multi-tenant authorization isolation.
- ✅ **Admin Review Moderation:** Spam flagging and automatic exclusion from product rating calculations.
- ✅ **Product Q&A Lifecycle** : Asking questions, upvoting questions, official vendor answering, and verified seller badging.